Laspeyre's price index number

$$P_{01}^{L}=\frac{\sum p_1 q_0}{\sum p_0 q_0}\times 100=\frac{1110}{952}\times 100=116.60$$

On an average, there is an increase of 16.60 % in the price of the commodities when the year 2000 compared with the year 2010.

Paasche's price index number

$$P_{01}^{P}=\frac{\sum p_1 q_1}{\sum p_0 q_1}\times 100=\frac{1460}{1236}\times 100=118.12$$

On an average, there is an increase of 18.12 % in the price of the commodities when the year 2000 compared with the year 2010.

Fisher's price index number

$$P_{01}^{F}=\sqrt{\frac{\sum p_1q_0 \times \sum p_1q_1}{\sum p_0q_0 \times \sum p_0q_1}}\times 100$$

$$=\sqrt{\frac{1110\times 1460}{952\times 1236}}\times 100=117.36$$

On an average, there is an increase of 17.36 % in the price of the commodities when the year 2000 compared with the year 2010.

Example 9.11

Construct the Laspeyre's , Paasche's and Fisher's price index number for the following data. Comment on the result.

Commodities	Base Year		Current Year	
	Price	Quantity	Price	Quantity
Rice	15	5	16	8
Wheat	10	6	18	9
Rent	8	7	15	8
Fuel	9	5	12	6
Transport	11	4	11	7
Miscellaneous	16	6	15	10

Solution:

Commodities	Base Year		Current Year		p_0q_0	p_0q_1	p_1q_0	p_1q_1
	Price (p_0)	Quantity (p_0)	Price	Quantity				
Rice	15	5	16	8	75	120	80	128
Wheat	10	6	18	9	60	90	108	162
Rent	8	7	15	8	56	64	105	120
Fuel	9	5	12	6	45	54	60	72
Transport	11	4	11	7	44	77	44	77
Miscellaneous	16	6	15	10	96	160	90	150
				Total	**376**	**565**	**487**	**709**

Table 9.10

Laspeyre's price index number

$$P_{01}^{L}=\frac{\sum p_1 q_0}{\sum p_0 q_0}\times 100=\frac{487}{376}\times 100=129.5212$$

Paasche's price index number

$$P_{01}^{P}=\frac{\sum p_1 q_1}{\sum p_0 q_1}\times 100=\frac{709}{565}\times 100=125.4867$$

Fisher's price index number

$$P_{01}^{F}=\left(\sqrt{\frac{\sum p_1q_0 \times \sum p_1q_1}{\sum p_0q_0 \times \sum p_0q_1}}\right)\times 100$$

$$=\left(\sqrt{\frac{487\times 709}{376\times 565}}\right)\times 100=127.4879$$

On an average, there is an increase of 29.52 % , 25.48% and 27.48% in the price of the commodities by Laspeyre's, Paasche's, Fisher's price index number respectively, when the base year compared with the current year.

9.2.3 Test of adequacy for an Index Number

Index numbers are studied to know the relative changes in price and quantity for any two years compared. There are two tests which are used to test the adequacy for an index number. The two tests are as follows,

(i) Time Reversal Test

(ii) Factor Reversal Test

The criterion for a good index number is to satisfy the above two tests.

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